AI METHODOLOGY REPORT

ZILAL AL SAFA

How artificial intelligence shaped every stage of the design — with human-led decisions

Dubai Municipality
AI Park Design Challenge
Plot 3570616 · 15,000 m²
Concept + Preliminary Design

1 · Purpose & principle

The Challenge asks entrants to demonstrate how AI can be **meaningfully integrated throughout both the Concept and Preliminary design stages** — to support analysis, ideation, design development and decision-making — while maintaining human creativity and professional judgement. This report documents exactly that for ZILAL AL SAFA.

> **Our governing principle — "AI proposes, humans dispose."** At every stage, AI is used to **widen the option space** and to **quantify trade-offs** faster and more rigorously than manual methods allow. Every final spatial, cultural and aesthetic decision is made by the human design team. AI is a co-designer and analyst, never the author.

2 · The pipeline at a glance

AI is applied through a **six-stage**, reproducible pipeline. Uniquely, it does not stop at the drawing: a sixth stage carries AI into **live operation** of the built park. Each stage has defined inputs, tools, AI outputs, the alternatives evaluated, and the human decision that closed it.

Stage	Question it answers	Principal AI role
01 · Ingest	What is really happening on this site?	Data fusion, computer vision, NLP sentiment
02 · Generate	What layouts are worth considering?	Generative & parametric option search
03 · Simulate	How comfortable & usable is each option?	Environmental & pedestrian simulation
04 · Forecast	How will people actually use it?	Predictive experience & capacity models
05 · Visualise	How do we communicate it truthfully?	Diffusion-model imagery & data storytelling
06 · Operate	How does AI keep it comfortable after opening?	Live digital twin — the "Comfort Brain"

Every stage feeds the next; findings also loop back (e.g. simulation results re-seed a new round of generation). The loop is run until performance targets and design intent converge.

3 · Data foundation

The analysis is grounded in the supplied competition material and open data, not assumption:

Stage-by-stage method (1)

1 Ingest — site intelligence

Objective: build an evidence base of how the site performs and how people use it. **Tools/methods:** GIS & QGIS for spatial layers; computer-vision extraction of tree canopy, paths and hard/soft surfaces from aerial imagery; sun-path & wind-rose computation; NLP sentiment analysis of public reviews; parsing of the DWG survey.

AI outputs: a canopy & surface map, an hourly footfall curve, a heat/shade exposure baseline, and a ranked list of user-voiced issues (heat, shade, seating, facilities). **Human decision:** the team confirmed the core problem statement — “reclaim the midday heat-gap hours” — as the project's north star.

2 Generate — layout option search

Objective: explore many spatial arrangements rather than one. **Tools/methods:** parametric modelling (Grasshopper-style rule systems) plus generative models to vary path geometry, canopy placement, zone adjacencies and program mix, subject to constraints (retain pitch/court & mature trees, four gateways, ~1 km loop, ~15% leasable).

AI outputs: dozens of candidate layouts, each scored on shade-reachability, walkability (route directness & loop length), program adjacency and retained-asset preservation. **Human decision:** the team selected and refined the “Cool Loop wrapping a cool social core” family, rejecting options that fragmented the canopy or pushed active uses into the quiet core.

> **Representative generation brief (paraphrased):** “Generate park layouts on a 90×166 m plot that (a) provide a continuous shaded circuit ≥ ~1 km, (b) keep passive/social space in the thermally coolest zone, (c) retain the existing pitch, court and mature trees, (d) give each of the four edges a distinct entrance, and (e) reserve ~15% for leasable commercial use. Rank by shade-reachability and route directness.”

Stage-by-stage method (2)

3 Simulate — comfort & flow

Objective: test how each shortlisted option actually performs. **Tools/methods:** hourly/seasonal shade ray-tracing and UTCI thermal-comfort modelling (Ladybug/Honeybee-type environmental tools) driven by Dubai climate data; pedestrian-flow modelling across the path network at the 6 pm peak.

AI outputs: per-hour shade maps, comfortable-area percentages (e.g. comfortable area at 3 pm July rising from ~9% to ~63%), and congestion/desire-line diagnostics. **Human decision:** canopies were concentrated on ray-traced hot-spots (“more comfort per dirham”) and the active edge was oriented toward the busy arrival, keeping the grove calm.

4 Forecast — experience & capacity

Objective: anticipate how the redesigned park will be used before it is built. **Tools/methods:** machine-learning models estimating dwell time, hourly capacity and inclusivity coverage by zone, calibrated to the Manual's targets (150–400 visitors / 10,000 m², 1–3 hour visits) and the observed footfall curve.

AI outputs: projected extension of comfortable usable hours (~+5.5 h/day in summer), balanced zone loading, and gap-detection for under-served user groups. **Human decision:** amenities (seating, WCs, water points, rest nodes ≤ 60 m apart) were distributed to remove the forecast gaps, and the events calendar was scheduled to the climate year.

5 Visualise — render & narrate

Objective: communicate the scheme truthfully and vividly. **Tools/methods:** diffusion-model image generation for day and night perspectives, art-directed by the team to match the plan; data-visualisation of the modelled results; assembly of the interactive microsite, board set and one-minute concept animation.

AI outputs: photorealistic concept perspectives (aerial, majlis, Cool Loop, all-abilities play, and a night view), the charts in this submission, and draft narrative text. **Human decision:** the team curated, corrected and captioned every image so visuals remain faithful to the measured design intent — renders illustrate the concept and are labelled as such.

Stage 6 — AI that operates the park

6 Operate — the Comfort Brain DUBAI-FIRST

Most AI park entries stop at the drawing. ZILAL carries AI into **live operation**, so the finished park stays comfortable in real time — our single strongest differentiator on the 20% AI criterion.

Objective: keep the built park comfortable, safe and efficient every hour.
Tools/methods: a live **digital twin** fed by on-site IoT sensors (temperature, UTCI, humidity, wind, air quality and footstep/footfall data, including the “Every Step Counts” kinetic tiles) applies the same UTCI logic used at design stage to the operating park.

AI outputs: real-time routing that guides visitors via the park app to the coolest available seat, lane or majlis; automated tuning of misting, irrigation and lighting to demand and weather; and an open KPI dashboard for Dubai Municipality. **Human decision:** operations staff set comfort/energy policies and thresholds; the AI executes and learns within them, improving each season.

The signature innovation stack

The Comfort Brain anchors four benchmarked signature innovations that make ZILAL unique on every side — each measured against real built parks (including Dubai Municipality's own Quranic Park) so every claim holds up:

Signature innovation	What it is & why it stands out
The Comfort Brain	Live AI digital twin that routes visitors to real-time comfort and auto-tunes the park — no Dubai park operates one.
Kinetic Mashrabiya canopy	Heritage screen reimagined as sun-tracking responsive shade over the Majlis Grove — iconic and unmistakably Emirati.
Net-positive solar canopy	PV spine shades the Cool Loop and generates more energy than the park uses — powering the tech and exporting surplus.
“Every Step Counts” kinetic floor	Footstep-energy tiles at the sports court & a Cool-Loop stretch; footfall doubles as a live sensor for the twin.

Costed honestly. The Comfort Brain and sensor network are low-cost, high-impact software + IoT. The one premium item — the footstep-energy floor — is deliberately limited to ~100–300 m² of hero zones and justified on **engagement, data and safety lighting rather than kWh**; the solar canopy carries the real energy load. All of it is sized to sit inside the AED 35M budget.

Coverage & toolchain

4 · AI applied across all required areas

The Scope lists specific areas in which AI should be applied. ZILAL uses AI in each:

Area of application	How AI was used in ZILAL
Site & context analysis	CV surface/canopy extraction, GIS fusion, DWG parsing
User & community analysis	Footfall curves, review-sentiment NLP, persona synthesis
Environmental analysis	Sun-path, shade ray-tracing, UTCI, wind modelling
Design ideation	Generative & parametric layout search
Design development & optimisation	Iterative scoring of shade, walkability, adjacency
Sustainability & resilience	Irrigation/water-balance & solar-yield estimation
User experience & decision-making	Dwell-time/capacity forecasting, gap detection
Visualisation & communication	Diffusion renders, data viz, animation, microsite
Live park operation	Real-time digital twin (“Comfort Brain”): routing, auto-tuning & a KPI dashboard

5 · Toolchain summary

Function	Representative tools / techniques	Output used
Spatial data & GIS	QGIS, satellite/aerial imagery, DWG survey	Base plan, context & surface maps
Computer vision	Segmentation of canopy / paths / surfaces	Existing-conditions baseline
Behaviour & NLP	Popular-times data, review-sentiment analysis	Footfall curve, issue ranking, personas
Generative design	Parametric (Grasshopper-style) + generative models	Layout option set & scoring
Environmental sim	Ladybug/Honeybee-type shade & UTCI, wind	Shade & comfort maps, KPIs
Predictive modelling	ML dwell-time / capacity / coverage	Usage & inclusivity forecasts
Visualisation	Diffusion image models; charting; web	Renders, charts, microsite, animation

Tool names are representative of the methods applied at concept stage; equivalent industry-standard tools may be substituted during detailed design without changing the methodology.

6 · Validation, limitations & assumptions

- **Modelled, not measured:** figures are concept-level estimates grounded in Manual targets, climate normals and public data. They will be validated with an on-site survey, metered baselines and detailed simulation in the next stage.
- **Human validation:** every AI output was reviewed by the design team; implausible or off-brief results were discarded, not published.
- **Reproducibility:** inputs, constraints and scoring criteria are documented so the pipeline can be re-run and audited.

7 · Responsible-AI governance

- **Human accountability:** named designers own every final decision; AI outputs are advisory.
- **Bias & inclusivity:** forecasts were explicitly checked for under-served groups (People of Determination, seniors, children, women, teenagers) and gaps were designed out.
- **Data ethics:** only aggregate, non-personal, publicly available behavioural data was used.
- **Transparency:** AI-generated imagery is labelled as concept visualisation; no render is presented as a photograph of an existing condition.

> **Human creativity + AI rigour.** The cultural and creative core of ZILAL — the Ghaf-tree identity, the majlis social heart, the “reclaim the lost hours” idea, the sense of place — is human. AI's contribution is to make that vision **measurably better and defensible**: to prove the shade works at 3 pm in July, that the loop is genuinely ~1 km and walkable, that the water and energy numbers add up, and that no user group is left out.

Why this wins the AI criterion

8 · The differentiator

Most entrants will submit renders and a paragraph about AI. ZILAL submits a **documented, reproducible pipeline** where AI demonstrably shaped analysis, ideation, development, decision-making and communication — with human judgement retained throughout — and then goes one decisive step further: AI also **operates** the finished park through the live **Comfort Brain**. This is our strongest Dubai-first, and it is precisely what the "Integration & Effective Use of AI" criterion (20% of the score) rewards.

The Comfort Brain anchors a stack of four benchmarked signature innovations — the AI comfort twin, the sun-tracking **kinetic mashrabiya** canopy, a **net-positive solar** canopy, and the interactive **"Every Step Counts"** kinetic floor whose footfall doubles as a live sensor for the twin — making the park not only AI-designed but genuinely interactive and unique on every side.

9 · From concept to preliminary design

The same pipeline carries into the Preliminary stage: generation is narrowed to detailed layout resolution; simulation is re-run at higher fidelity on the chosen scheme; forecasting informs amenity sizing and phasing; and visualisation produces the developed drawings, sections, boards and animation accompanying this submission.

Stage	Concept design use	Preliminary design use
Ingest	Problem framing & baseline	Survey-calibrated data
Generate	Broad layout families	Resolved layout & details
Simulate	Comparative comfort screening	High-fidelity verification
Forecast	Usage direction	Amenity sizing & phasing
Visualise	Concept renders & microsite	Developed drawings, boards, animation
Operate	Comfort Brain concept & sensor plan	Commissioned digital twin & live dashboard

Summary: AI is not a garnish on ZILAL — it is the method by which a culturally-rooted, human-authored vision was made comfortable, inclusive, sustainable and buildable, and proven with numbers.

This report accompanies the ZILAL AL SAFA concept + preliminary design submission (interactive microsite, drawing set, board set, renders, data charts and one-minute animation).